1. Draw a function that is continuous everywhere except for $x = -2, 1, \& 5$. Use a different type of discontinuity at each of these values of *x*.

2. Determine if the following function is continuous at $x = -2$. $f(x) = (x+1)^2$.

3. Is the following piecewise function continuous at $x = -2$?

$$f(x) = \begin{cases} (x+1)^2, & x \neq -2 \\ 1, & x = -2 \end{cases}$$

4. Is the following piecewise function continuous at $x = -2$.

$$f(x) = \begin{cases} \dfrac{x^2 - x - 6}{x^2 + 5x + 6}, & x \leq 0 \\ \dfrac{\ln(x+3)}{e^{x+2}}, & x > 0 \end{cases}$$

5. Is the following function continuous at $x = -2$? Justify your answer.

$$f(x) = \begin{cases} \dfrac{x^2 - x - 6}{x^2 + 5x + 6}, & x \leq -2 \\ \dfrac{\ln(x+3)}{e^{x+2}}, & x > -2 \end{cases}$$

6. Where is the function $y = \dfrac{e^{\frac{1}{x-1}} + \sin(3x^2 + 2)}{\tan(x)}$ continuous?

7. For what value of the constant *c* is the function *f* continuous for all real numbers?

$$f(x) = \begin{cases} cx^2 + 5x & if \quad x < 3 \\ x^3 - cx & if \quad x \geq 3 \end{cases}$$

8. Consider the function $f(x) = \begin{cases} \dfrac{x^2 - 9}{x - 3} & x < 3 \\ ax - 1 & x \geq 3 \end{cases}$.

a) For what value of *a* is the function continuous at $x = 3$?

b) If $a = 5$, what type of discontinuity exists at $x = 3$? Justify your answer.

The following questions are related to the Intermediate Value Theorem (IVT)

9. Something is wrong with the following 'proof'. What is it?

   Let $f(x)=\frac{x}{|x|}$. When $x=1$, $f(x)=1$. When $x=-1$, $f(x)=-1$. There, by the IVT, $f(c)=0$ for some c between -1 and 1.

10. Let C be the ellipse $x^2+4y^2=1$. How can one use the IVT to show that there is point P on C such that the area of the triangle determined by $(1,0)$, $(-1,0)$ and C is $\frac{\pi}{12}$. Hint: How does the area of the triangle change as one chooses different points on C?

11. Show that there exists a rectangular box with edges of length x, $x+1$, and $x+2$ for some positive value x so that the volume of the box is equal to its surface area (assume the box has both a top and a bottom).